

Status of Fluorosis in Animals

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Abstract Excess intake of fluoride for a prolonged period results in its accumulation predominantly in hard tissues, teeth and bones and causes diverse adverse effects that appear in the form of dental and skeletal fluorosis. Prolonged ingestion of fluoride also affects other organs and alters their architecture and physiology. Dental, skeletal and non-skeletal fluorosis have been described in different species of domestic animals due to excess concentration of fluoride in their drinking water, but maximum permissible or safe limit of fluoride in drinking water for these animals has not been well studied. Effects of different levels of fluoride exposures have been studied experimentally in laboratory animals, including rabbit, rat and mice. It has been found to affect digestive tract, liver, kidneys, brain, and endocrine organs, besides intrauterine development of foetus. But the exact mechanism involved in fluoride toxicity in these organs or tissues remain unanswered. Amelioration of fluorototoxicosis has been well observed in soft tissues in experimental study by using certain ameliorative agents such as ascorbic acid, vitamin A and D, calcium, boron, magnesium, and selenium compounds. Some herbal amelioration has also been investigated recently in sheep, goats and camels. This review provides comprehensive information on status of fluorosis in domestic and laboratory animals.

Keywords Amelioration · Apoptosis · Excitotoxicity · Fluorosis · Genotoxicosis · Neurotoxicosis · Reproductive and teratogenic effects

Introduction

Fluorine, the ninth element of the periodic table was discovered by Professor Henry Moissan on June 26, 1886. However, its application and biological significance came to be known very late in the decade of 1920. Owing to its electronic configuration ($1s^2 2s^2 2p^5$), fluorine is the most electronegative and highly reactive diatomic pale yellow-green gas, hence, not found free in the nature in element form. It has strong affinity to combine chemically with other elements to form compounds called ‘fluorides’. Many researchers and reviewers use the word fluoride to denote the ionised, physiologically available form of the element. The word ‘fluorine’ has sometimes been thought to denote exclusively the gaseous form of the element. However, by analogy with iodine, fluorine may be used to denote collectively the element in all its forms, ionised or ionisable. Where, there is doubt as regards to ionisability, e.g. in certain food, fluorine is actually, more comprehensive [1].

Fluoride is varyingly distributed in sea water, drinking water, soil, dust and in mineral deposits of fluor spar, cryolite, fluoroapatite, etc. The principal sources of fluoride to man and animals are drinking water, vegetation grown on fluorotic soils and water, certain edible marine animals, dust in air, and certain industrial processes [1]. However, drinking water is the main source of chronic fluoride intoxication to man and animals.

Fluoride Distribution

Natural waters having as high as 2,800 ppm fluoride concentration have been reported from different parts of the world [1]. In India, except a few eastern states high fluoride content is reported in ground water of different states [1–4] including Andhra Pradesh (1.02–13.0 ppm), Arunachal

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Pradesh (0.1–0.44 ppm), Assam (0.1–0.21 ppm), Bihar (0.1–0.2 ppm), Chhattisgarh (1.0–8.8 ppm), Delhi (Union territory) (0.4–3.93 ppm), Gujarat (0.2–20.0 ppm), Haryana (0.17–24.7 ppm), Himachal Pradesh (0.0–1.0 ppm), Jammu and Kashmir (0.05–4.12 ppm), Karnataka (0.1–18.0 ppm), Kerala (0.0–0.05 ppm), Madhya Pradesh (0.2–7.6 ppm), Maharashtra (0.11–10.21 ppm), Orissa (0.1–52.5 ppm), Punjab (0.3–16.2 ppm), Rajasthan (0.12–90.0 ppm), Tamil Nadu (0.5–4.0 ppm), Uttar Pradesh (0.1–19.0 ppm) and West Bengal (0.39–6.80 ppm). In these states, 50–100 % districts are affected with fluoride in Andhra Pradesh, Gujarat, Rajasthan and Uttar Pradesh, while in Bihar, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Punjab and Tamil Nadu, 30–50 % districts are endemic for fluoride [2]. Despite the occurrence of high fluoride content in ground water of these states studies on its toxicosis in diverse species of domestic animals are still meagre [5–7].

Fluorotoxicosis (Fluorosis)

Biologically, fluoride is regarded both beneficial and harmful to man and animals. It is required for about four important physiological functions: calcification of teeth and bones, fertility maintenance, hematopoiesis, and activation of certain enzyme systems hence, it has been considered as one of the 14 essential elements for animal life [8]. But the necessity of fluoride in biological functions is still doubted, probably because the amount required is so small that deficiency can not be produced under even the most careful laboratory conditions using trace element sterile isolators [9]. Readily soluble fluoride compounds are relatively more toxic and are easily absorbed by digestive tract of animals. In general, inorganic forms of fluoride are more toxic than its organic forms [10].

Significance of fluoride in veterinary medicine is mostly due to its excessive intake because ingestion of small amounts does not cause any severe illness [10]. But due to accumulation of fluoride in biological systems, constant and increasing intake results in pathognomonic signs of intoxication. The condition is referred to as fluorotoxicity or fluorosis [9]. In general, these signs appear in teeth, bones and soft tissues or organs of body and are known as dental, skeletal and non-skeletal fluorosis, respectively.

The most recognisable, indexive and early sign of chronic fluoride intoxication is dental fluorosis, which is characterised by light yellow to brown or black horizontal streaks (hypoplasia), pitches and fine dots on the enamel surface of teeth causing loss of natural glossy shining and early loss of the teeth. Skeletal fluorosis is more dangerous and highly significant since it diminishes the mobility of animals in very early age by producing varying changes in the bones such as exostosis, osteosclerosis, osteoporosis,

osteophytosis and calcification of interosseous membranes, ligaments and tendons [10, 11]. These changes appear clinically in the form of vague aches and pains in the body and joints which are associated with rigidity and lameness, palpable bony lesions, snapping sound in feet during walking, emaciation and intermittent diarrhoea [12, 13].

Sources and Tolerance of Fluoride

Domestic animals viz., cattle, buffaloes, sheep, goats, camels, horses, donkeys, etc. get excessive fluoride mainly from fluoridated drinking water, plants growing on fluoride containing soils, fluoride bearing industrial effluents, dust, and wastes and fluoride-rich feed supplements. The first two sources are natural and generally responsible for endemic fluorosis, whereas the other two sources are man-made hence restricted to a particular location.

Domestic animals have a threshold level of fluoride tolerance, above which fluorotoxicosis develops. Young animals are more prone to fluoride deposition in body tissues. Fluoride toxicity is reported to be dependent on various factors as species, age, food, nutrients, fluoride exposure, intake frequency, ambient factors, etc. [6, 7]. This leads to difficulty in setting a rigid tolerance limit of fluoride for animals. Erstwhile USSR recommended a limit of 1.0 mg/m³, while W.H.O. standardized 1.5 ppm in drinking water [14]. Two general criteria have been used to define dietary tolerance of fluoride for farm animals. The level of fluoride, which if fed continuously would not interfere with the productive ability of animals is termed as performance level, and that above which some pathological changes develop, is known as pathology level. The pathology level, however, is an arbitrary value and fluoride induced dental lesions may be detected at almost any levels of fluoride above the normal 5 ppm [15, 16]. Recent studies conducted in low and high fluoride endemic areas revealed that osteo-dental fluorosis in diverse species of domestic animals can appear at >1 ppm fluoride concentration in drinking water [17–21].

Epidemiological Investigations

Most of the work on fluorotoxicosis has been conducted on dairy animals exposed with mineral supplement containing soft rock phosphate and protein supplement containing high level of fluoride [11, 22–24] and in small animals (rat, mice and rabbits) exposed experimentally with fluoridated drinking water having different concentrations of fluoride (NaF). Some data on industrial fluorosis in animals is also available [10]. However, investigations on natural chronic fluoride intoxication in domestic animals inhabiting different fluoride endemic areas are still too scanty. Although many states are hyper-endemic for hydrofluorosis [1–4],

nevertheless, a few epidemiological studies on natural hydrofluorosis in different species of animals living in the same fluoride endemic areas have been conducted in India [17–21]. Findings reveal that prevalence and severity of osteo-dental fluorosis increases with increasing age, amount of fluoride, its duration of exposure and intake frequency. Other biological and non-biological determinants that accelerate fluoride toxicity have also been discussed [6, 7] and it has been observed that different species of domestic animals are almost equally susceptible to fluoride toxicity, but its severity varies greatly due to different nutrients in their food and feed [25–27]. Osteo-dental fluorosis (Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10) occurs in these species (*Bos taurus*, *Bubalus bubalis*, *Ovis aries*, *Capra hircus*, *Equus caballus*, *E. asinus* and *Camelus dromedarius*) at above 1 ppm fluoride concentration in drinking water [26]. However, more epidemiological studies are needed in different domestic animal species

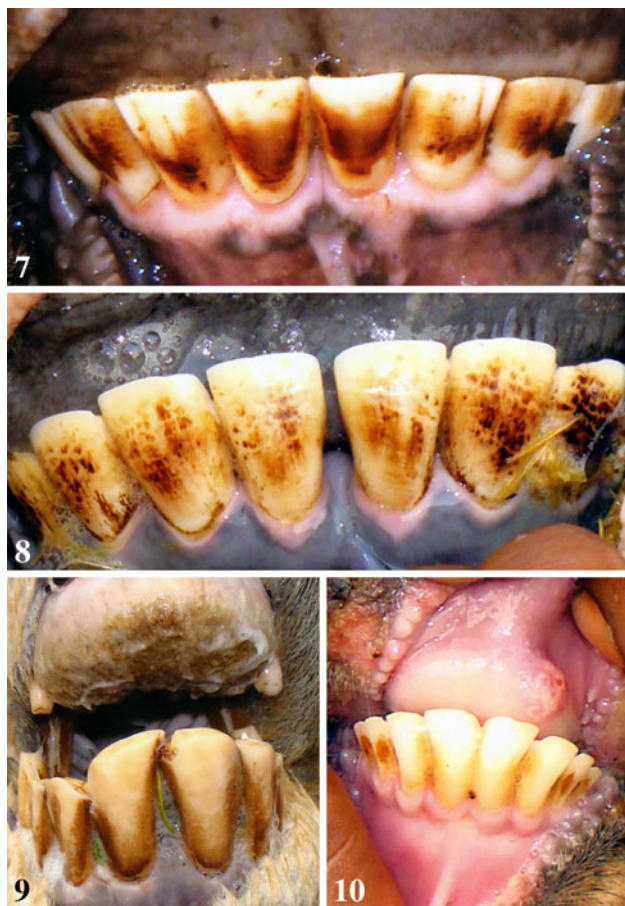
living in same fluoride endemic areas for the assessment of fluoride tolerance or susceptibility.

Fluorotoxicosis has also been found in animals grazing pastures near fluoride emitting industries such as phosphate mining and fertilizer plants [28–32]. A number of industrial fluorosis outbreaks have been associated with environmental pollution from aluminium industries. A report indicated that cattle grazing within eight miles of aluminium plants could develop signs of fluoride toxicity [10]. However, industrial fluorosis is restricted to a particular locality.

Skeletal changes have been confirmed radiologically (X-ray) by a few workers [33]. However, data on radiological changes in animals affected with chronic fluoride intoxication are meagre. Hence more radiological studies are needed to confirm these adverse effects of fluoride intoxication. Fluoride is also reported to affect growth and milk production [11, 34] but these findings are still controversial and require more investigations.

Figs. 1–6 Calves of buffalo (1) and cattle (3) under 2 months of age exhibiting severe dental fluorosis, hind leg lameness, intermittent diarrhoea, and wasting of body muscles as signs of skeletal fluorosis. Mature sheep showing moderate dental (5) and skeletal fluorosis (6). Source: Choubisa et al. [26]





Figs. 7–10 Severe dental fluorosis in a mature cow (7) and a buffalo (8). Moderate dental fluorosis in a camel (9) and a goat (10) living in areas with 1.5–1.7 ppm F in their drinking waters. Source: Choubisa et al. [26]

Effect on Digestive Tract, Liver and Spleen

Fluoride is directly absorbed from different regions of alimentary canal via passive diffusion mechanism [35, 36] and reaches different body parts through blood circulation. Acute and chronic investigations in animals have shown that NaF causes gastro-intestinal damage [37, 38]. Gastro-intestinal manifestations including abdominal pain, constipation and intermittent diarrhoea are found in animals living in fluoride endemic areas [19, 20]. Fluoride induced histopathological changes in the duodenum included surface abrasion of mucosa, clumped sub mucosa, diffuse punctuate haemorrhage, necrosis of crypt cells and Brunner's glands and hypertrophy of musculature in experimental animals exposed to different concentrations of NaF [39, 40]. These changes are attributed to rapid penetration of hydrofluoric acid across the gastro-intestinal cell membranes [40]. But exact mechanism involved in histopathological alterations in the intestinal tissues is not clear.

Like many other stressogenic agents, fluoride enhances metabolic rate and hepatic accumulation of ascorbic acid and lipids mainly triglycerides in animals [41, 42]. Centrilobular necrosis and mononuclear filtration in the portal triad area are evident in the liver of calves that were fed diet supplemented with fluoride [43].

Among the major lymphoid organs of the body, spleen plays a vital role in hemopoiesis especially during foetal life. In mice it is actively involved in blood cell production throughout the life [44]. Although spleen is an important organ but studies on fluoride toxicity on this organ are limited. Recently, a report on this subject stated that after administration of NaF in low (15 ppm) and high (150 ppm) doses through drinking water for 90 days in Swiss albino mice, a decrease in the organo-somatic index (OSI) accompanied by a decline in the white pulp content of the spleen [45] was observed.

Renal Effects

Among all the organs kidneys are more susceptible and sensitive to fluoride depositions [10]. Therefore, exposure of excessive amounts of fluoride causes both structural and functional damage to kidney [46, 47]. Intoxication with high fluoride content (>190 ppm) in drinking water induced renal injury in experimental animals such as tubular degeneration, inflammation, fibrosis, parenchymatous nephritis, cloudy swelling, dilation of convoluted tubules and apoptosis [48, 49]. All these changes in the kidney result in impaired renal function. Kidney tissues of these animals revealed increased activity of lactate dehydrogenase (LDH) and decreased activity of alkaline phosphatase. Serum of these animals also showed increased level of urea, nitrogen and creatinine but decreased Na^+ . An inhibition of various enzyme systems in kidney and decrease in renal Na^+ , K^+ and ATPase activities due to action of fluoride on kidneys [10, 50] has also been seen. Elevated serum urea, nitrogen and creatinine are reported in cows and buffaloes afflicted with fluorosis [34]. These findings indicate deleterious effects of excessive fluoride exposure on kidneys. Recently an in vitro study revealed that at the lowest concentration of 20 $\mu\text{mol NaF/l}$, hormesis or stimulatory effect of NaF occurred on growth of kidney cells [51].

Neurotoxicosis

The mechanism of the neurotoxic effect of fluoride and alumina fluoride complex on the brain has not been fully elucidated, although there is compelling evidence that it is closely related to that of heavy metal neurotoxicities as well as a host of neuropathological conditions. Cattle, buffaloes and other domestic animals living in

hyper-endemic areas for long time develop para and quadriplegia, polyurea and neurological disorders [6, 7]. Fluoride shows the ability to cross the blood–brain barrier [52]. Fluoride accumulation was observed in the brain of experimental animals exposed to chronic high fluoride intake and this accumulation increased with the increase of fluoride level in drinking water [53]. Chronic fluoride exposure also alters the neuronal and cerebro vascular integrity, behaviour patterns and induces metabolic lesions in the brain [54, 55] as observed in studies done on experimental animals [56, 57]. These changes are reported to be initially due to lipid peroxidation caused by interaction of high fluoride and low iodine and apparently occurred mainly during the period of embryonic development and in the early stage of brain development.

Apoptosis

Apoptosis, the natural process of programmed cell death can be induced by chronic fluoride exposure, which has been observed and reported in lung, kidney, liver and bone tissues [58–61]. Fluoride induced apoptosis is not only pronounced in the periphery, but is also extensive in the central nervous system. In an experiment on rats, intraperitoneal injection of fluoride (20 ppm NaF/Kg body weight) caused apoptosis/cell death. Flow cytometry (FCM) indicated that the percentages of apoptotic cell death in brain cortex and hippocampus were significantly higher ($P < 0.01$) in rats after giving this treatment [62]. The same results were also found for neuron apoptosis in adult and offspring rats exposed with high fluoride (100 ppm NaF) in drinking water [63, 64].

Excitotoxicity

Recently evidences indicate that fluoride induces neural destruction and synaptic injury by a mechanism that involves free radical production and lipid peroxidation in the hippocampus of brain [56, 65]. Glutamate is the most abundant neurotransmitter in the brain and operates via a series of specific receptors, either ionotropic or G-protein-operated metabotropic receptors. When these receptors are over stimulated, a series of intracellular signalling events occur that can lead to cell death, a process called excitotoxicity [66]. The exact mechanism of excitotoxicity is not yet clear. However, possible control mechanism of excitotoxicity or fluoride neurotoxicity has been proposed [67]. The most common mechanism involved in excitotoxicity is one free radical generation and inhibition of antioxidant enzymes, mitochondrial energy enzymes, and of glutamate transporters [66, 67]. But none of these mechanisms as yet has actually looked at it in relation to fluoride neurotoxicity, despite the strong evidence that

fluoride can accumulate in certain areas of the brain in relatively high concentration [67].

Histopathological studies of fluoride exposed animals have demonstrated damage to CA 1 and CA 4 areas of the hippocampus and to the dentate gyrus, due to excitotoxicity. Using NaF and AlF_3 damage in the superficial layers of the cortex, amygdale and cerebellum—all areas endowed with abundant glutamate receptors have been observed [54].

Reproductive and Teratogenic Effects

Chronic fluoride intoxication can impair reproductive function. High prevalence of sterility, still birth, repeated oestrus cycle, and birth of weak offspring have been reported in domestic animals in the areas with high fluoride contents in water. The male animals show poor libido and females show low conception rate [19, 20]. Cows exposed with fluoride (8–12 ppm/kg/day) for a year, increased post-calving anoestrus and declined in fertility [10] while in fluoride treated rabbits and rodents, structural and physiological defects in spermatozoa, a decrease in sperm count, and alterations in the epididymis and accessory reproductive glands, reduced fertility and disturbances in the levels of reproductive hormones have been reported [68–74]. However, the exact mechanism or molecular pathways of fluoride effects on reproductive function is still not clear and reported findings are conflicting. Tao and Suttie [75] refuted any adverse role of fluoride in mammalian reproduction. Another laboratory study [76] indicated no adverse effect of fluoride on spermatogenesis or endocrine function at 25, 100, 175 and 250 ppm fluoride in drinking water of male rats while, Cui et al. [77] reported that 150 ppm fluoride for 10 weeks in drinking water of rats caused significant decrease in the sperm count and sperm mobility.

Many studies suggest that placental transfer of fluoride does occur in animals, and that it may exert effects on developing foetus causing fetotoxicity. In experimental studies, radiographs (X-rays) of the foetuses of fluoride treated rats revealed retardation of skeletal development, delayed ossification and shortening of bone elements [78]. In fluoride endemic areas calves of fluorosed cattle and buffaloes showed abnormal morphology which indicates placental transfer of fluoride [21].

Effects on Endocrines

Extremely high intake of fluoride may influence function of thyroid, parathyroid and adrenal gland and cause subsequent changes in humoral profile [9, 10]. Although precise mechanism of the effects of fluoride on complex thyroid cycle is not yet clear, it is opined that fluoride competes with

iodine when thyroxine is synthesised. The hypothesis states that fluoride inhibits the proteinase responsible for splitting thyroglobulin molecule into T_3 and T_4 [79]. Thus animals subjected to excess feeding of fluoride may show hypothyroidism and decreased T_3 and T_4 levels which may increase after administration of iodised casein [80]. Decreased mean values of serum T_3 (80.14 ng/dl) and T_4 (4.11 ng/dl) have been reported in fluorotic sheep than ($T_3 = 136.48$ ng/dl and $T_4 = 5.61$ ng/dl) normal sheep [81]. However, no significant change in the function of the thyroid gland was found in fluorosed cattle [10]. In a recent study, decreased level of plasma free thyroxine (FT_4), free tri-iodothyronine (FT_3) and reduction in body weight was observed in 14-day-old babies of adult female mice exposed with 226 ppm fluoride in their drinking water. Reduction in body weight (15 %) attributed to a defect in plasma thyroid hormone levels and hypertrophy of thyroid gland due to increased TSH level [82]. These changes reverted to normal when fluoride was eliminated from the drinking water. Development of hypo-thyroidism in cattle [80], calves [83], adult rats [84] and both adult [85] and young mice [86, 87] have been observed after ingestion of fluoride.

Parathormone is the most significant regulator of calcium homeostasis. Any situation that causes decrease in circulating calcium would induce parathormone release and if prolonged, would lead to secondary hyperparathyroidism [10]. In animals, studies on parathyroid activities in relation to fluoride intoxication are too scanty. However, elevated serum immunoreactive parathormone (IRPTH) was found in young sheep reared on water containing 100 ppm fluoride and also those given daily doses of fluoride for a week [10].

Weight of adrenal gland, plasma epinephrine and blood glucose was reported to increase significantly in animals afflicted with fluoride intoxication [88]. These changes are possibly associated with general stress response; secondary to the fluoride induced metabolic alterations as cyto-architecture did not exhibit any specific localised lesion. However, hypertrophy of adrenal cortex and medulla cells was noted [88]. High fluoride concentration also affects pituitary function and inhibits the action of trophic hormones of the hypophysis, thereby affecting growth and reproduction [10].

Genotoxic Effects

Genotoxicity due to fluoride intoxication has been reported mostly in laboratory animals including rabbit, mice, rats, etc. Fluoride induced DNA damage and breakage of DNA strands in these animals has also been observed [89, 90]. These studies reflected that the action of fluoride in vivo is actually more genotoxic at certain lower concentrations than at higher concentrations. Some other studies also

indicated chromosome aberrations, induction of sister chromatid exchange and DNA damage due to fluoride [91, 92]. Depletion of ovarian DNA and RNA content was observed in rabbits exposed to NaF subcutaneously in daily doses of 5, 10, 20 and 50 ppm/kg body weight for three and a half months [93]. Recent study on genotoxicity was conducted in freshwater cat fish (*Clarias batrachus*) exposed to sub lethal concentrations of fluoride (35 and 70 ppm in water). After 90 days of fluoride exposure kidney cells of both groups of these fishes showed evidence of genotoxic alterations that included chromatid deletions, acentric fragments, ring chromosomes and fragments [94].

Amelioration of Fluorotoxicosis

Fluoride, above the threshold level in body causes histological, histochemical and physiological changes in soft and hard tissues. But some salts of boron, calcium, magnesium, selenium, molybdenum and aluminium have ameliorative effect on fluoride toxicity and are capable of reducing the gastrointestinal absorption of fluoride [1, 10]. The toxic effects of fluoride in soft tissues of different organ systems can be reversed or normalised after withdrawal of fluoride exposure. These changes can also be reversed or mitigated in experimental fluorotic animals by giving a supplement of vitamins and mineral antioxidants such as vitamins A, C and E, and selenium [1, 95].

Interestingly, an amelioration of fluoride effects has been observed and reported in ruminants [25–27]. An extensive survey was carried out for the prevalence and severity of dental and skeletal fluorosis in different domestic animal species living in same fluoride endemic areas having more than 1 ppm fluoride in drinking water. The prevalence and severity of fluorosis in sheep, goats and camels were found relatively less as compared to cattle and buffaloes (Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10). Perhaps intake of fresh leaves, fruits and pods by these animals provide sufficient Ca and vitamin C to counteract the fluoride toxicity [25–27]. Mitigation of fluorotoxicosis has also been reported in dogs, rats and mice and penguins by ingestion of tamarind, black tea infusion and chitin, respectively [96–99].

Conclusion

Most of the studies on fluoride toxicosis have been conducted in small animals by giving high doses of fluoride in the form of NaF in drinking water and various pathological changes in digestive tract, liver, spleen, kidney, gonads, brain, chromosomes, intra-uterine development, etc. have been reported. Studies regarding the possibility of these changes in different species of domestic animals that are

naturally afflicted to chronic fluoride intoxication are highly suggestive. As these animals are economically important, more epidemiological studies are needed on natural fluorotoxicosis or hydrofluorosis in relation to different fluoride concentrations in drinking water and biological and non-biological determinants in diverse species of domestic animals to determine safe limit of fluoride in drinking water. In addition to this more studies are also required on molecular, histochemical, histoenzymological, biochemical and radiological aspects so as to assure the mechanism involved in fluoride toxicosis.

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References

- Sompura K (1997) Study on prevalence and severity of chronic fluoride intoxication in relation to certain determinants of fluorosis. PhD Thesis, Mohanlal Sukhadia University, Udaipur, Rajasthan
- Susheela AK (1993) Prevention and control of fluorosis in India, health aspects. Rajiv Gandhi National Drinking Water Mission, Ministry of Rural Development, New Delhi, pp 20–23
- Choubisa SL (2001) Endemic fluorosis in Rajasthan. Indian J Environ Health 42(4):177–189
- Choubisa SL (2001) Endemic fluorosis in southern Rajasthan. Fluoride 34(1):61–70
- Dwivedi SK, Dey S, Swarup D (1997) Hydrofluorosis in water buffalo (*Bubalus bubalis*) in India. Sci Total Environ 207(2–3):105–109
- Choubisa SL (1999) Some observations on endemic fluorosis in domestic animals in southern Rajasthan (India). Vet Res Commun 23:457–465
- Choubisa SL (2007) Fluoridated ground water and its toxic effect on domesticated animals residing in rural tribal areas of Rajasthan (India). Int J Environ Stud 64:151–159
- Anonymous (1973) Trace element in human nutrition. A report of a WHO expert group. Technical Series No. 532, World Health Organization, Geneva
- Wheeler SM, Fell LR (1983) Fluorides in cattle nutrition. Nut Abst Rev-B 53:741–767
- Swarup D, Dwivedi SK (2002) Environmental pollution and effect of lead and fluoride on animal health. Indian Council of Agricultural Research, New Delhi, pp 68–106
- Singh JL, Swarup D (1995) Clinical observation and diagnosis of fluorosis in dairy cows and buffaloes. A case report. Agr Pract 16:25–30
- Choubisa SL, Pandya H, Choubisa DK (1996) Osteo-dental fluorosis in bovines of tribal region in Dungarpur (Raj). J Environ Bio 17:85–92
- Choubisa SL (1999) Chronic fluoride intoxication (fluorosis) in tribes and their domestic animals. Int J Environ Stud 56:703–716
- Anonymous (1984) Environmental health criteria. 36: fluorine and fluorides. World Health Organization, Geneva, p 136
- Suttie JW (1980) Nutritional aspects of fluoride toxicosis. J Anim Sci 51:759–766
- Shupe JL, Olson MS, Peterson HB, Low JB (1984) Fluoride toxicosis in wild ungulates. J Am Vet Med Assoc 185:1295–1300
- Choubisa SL (1996) An epidemiological study on endemic fluorosis in tribal area of southern Rajasthan. A technical report. Ministry of Environment and Forests, Government of India, New Delhi, pp 1–86
- Choubisa SL (2008) Dental fluorosis in domestic animals. Curr Sci 95:1674–1675
- Choubisa SL (2010) Osteo-dental fluorosis in horses and donkeys of Rajasthan (India). Fluoride 43:5–12
- Choubisa SL (2010) Fluorosis in dromedary camels in Rajasthan (India). Fluoride 43:194–199
- Choubisa SL, Mishra GV, Sheikh Z, Bhardwaj B, Mali P, Jaroli VJ (2011) Toxic effects of fluoride in domestic animals. Adv Pharmacol Toxicol 12:29–37
- Clark RG, Hunter AC, Stewart DJ (1976) Deaths in cattle is suggestive of sub acute fluorine poisoning following ingestion of super phosphate. New Zealand Vet J 24:193–197
- Shlosberg A, Bartana U, Egyed MN (1980) Fluorosis in cattle due to high fluoride rock phosphate supplements. Fluoride 13:57–64
- Singh JL, Swarup D, Dogra RKS, Singh GR (1995) Fluorosis in dairy cows: clinicopathological findings. Indian J Anim Sci 65:162–165
- Choubisa SL (2010) Natural amelioration of fluoride toxicity (Fluorosis) in goats and sheep. Curr Sci 99:1331–1332
- Choubisa SL, Mishra GV, Sheikh Z, Bhardwaj B, Mali P, Jaroli VJ (2011) Food, fluoride and fluorosis in domestic ruminants in the Dungarpur district, Rajasthan, India. Fluoride 43:69–70
- Sheikh Z (2011) Prevalence of fluorosis in domestic animals. Curr Sci 101(9):1124–1125
- Riet-Correa F, Oliveiru JA, Plendez MD, Schild AL (1983) Industrial pollution as a cause of fluorine poisoning near Rio Gerandi city, Brazil. Pesquisa Vet Brasile 13:107–114
- Swarup D, Singh YP (1989) Bovine fluorosis in brick kiln congested zone. Indian J Vet Med 9:12–14
- Samal UN, Naik BN (1992) The fluorosis problem in tropical sheep. Fluoride 25:183–190
- Swarup D, Dwivedi SK, Dey S, Ray SK (1998) Fluoride intoxication in bovines due to industrial pollution. Indian J Anim Sci 68:605–608
- Patra RC, Dwivedi SK, Bhardwaj B, Swarup D (2000) Industrial fluorosis in cattle and buffaloes in Udaipur, India. Sci Total Environ 253:145–150
- Kant V, Verma PK, Pankaj NK, Raina R (2009) Experimental osteo fluorosis in goats of Jammu & Kashmir. Vet Scan 4:1–4
- Suttie JW, Miller RF, Phillips PH (1987) Effects of dietary NaF on dairy cows. Effects on milk production. Dairy Sci 40:1485–1491
- Gharzouli K, Sentor A (1994) Fluoride absorption in vitro by the gastrointestinal tract of the rat. Fluoride 27:185–188
- Whitford G (1996) The Metabolism and toxicity of fluoride, 2nd edn. Karger Publishers, Basel, pp 17–22
- Shashi A (1999) Gastric lesion in experimental fluorosis. Asian J Microbiol Biotech Environ Sci 1:171–175
- Waldbott GL (1977) Gastric ulcer and fluoride. Fluoride 10:149–151
- Sandhi H, Gupta ML, Gupta GL (1995) Intestinal effects of sodium fluoride in swiss albino mice. Fluoride 26:21–24
- Easman RP, Steflike DE, Pashiey DH, McKinney RV, Whitford GM (1984) Surface changes in rat gastric mucosa induced by sodium fluoride. A scanning electron microscope study. J Oral Pathol 13:225–264
- Chinoy NJ (1991) Effects of fluoride on physiology of animals and human beings. Indian J Environ Toxicol 1:17–32
- Saralakumari D, Varadacharyulu NCH, Ramakrishna R (1988) Effect of fluoride toxicity on growth and lipids in liver, kidney and serum in rats. Arog J Health 15:24–29
- Kapoor V, Prasad T, Bhatia KC (1993) Effect of dietary fluorin on histopathological changes in calves. Fluoride 26:105–110

44. Kotoky P, Barooah PK, Baruah MK et al (2008) Fluoride and endemic fluorosis in the Kabiangiong district, Assam, India. *Fluoride* 41:72–75
45. Podder S, Chattopadhyay A, Bhattacharya S, Ray MR (2010) Histopathological and cell cycle alteration in the spleen of mice from low and high doses of sodium fluoride. *Fluoride* 43:237–245
46. Shupe JL, Bruner RH, Seymour J, Aiden CL (1992) The pathology of chronic bovine fluorosis. *Toxicol Pathol* 20:274–285
47. Shashi A, Singh JP, Thapar SP (2002) Toxic effects of fluoride on rabbit kidney. *Fluoride* 35:38–50
48. Kour K, Singh J (1980) Histological finding in kidneys of mice following sodium fluoride administration. *Fluoride* 13:163–167
49. Zhan X, Wang M, Xu Z, Li J (2006) Toxic effects on fluoride on kidney function and histological structure in young pigs. *Fluoride* 39:22–26
50. Suketa Y, Terui (1980) Adrenal function and changes in renal (Na & K-ATPase) activity of fluoride intoxicated rats. *Fluoride* 13:148–150
51. Tang Q, An X, Du J, Zheng Z, Zhou X (2008) In vitro homeosis effects of sodium fluoride on kidney cells of three day-old male rats. *Fluoride* 41:292–296
52. Geeraerts F, Gijs G, Finne E, Crokaert R (1986) Kinetics of fluoride penetration in liver and brain. *Fluoride* 19:108–112
53. Mullenix PJ, Denbesten PK, Schunior A, Kernan WJ (1995) Neurotoxicity of sodium fluoride in rats. *Neurotoxic Teratol* 17:169–177
54. Varner JA, Jensen KF, Horwarth W, Issacson RL (1998) Chronic administration of aluminium or sodium fluoride to rats in drinking water. Alterations in neuronal and cerebrovascular integrity. *Brain Res* 784:284–298
55. Vani ML, Reddy KP (2000) Effect of fluoride accumulation on some enzymes of brain and gastrocnemius muscle of mice. *Fluoride* 33:17–26
56. Shivarajashankar YM, Shivashankara AR, Bhat GP, Rao SH (2002) Brain lipid peroxidation and antioxidant systems of young rats in chronic fluoride intoxication. *Fluoride* 35:197–203
57. Ge YM, Wang JD, Ning HM, Wang SL (2005) Effect of high fluoride and low iodine on brain histopathology in offspring rats. *Fluoride* 38:127–132
58. Refsnes M, Kersten H, Schwarze PE, Lag M (2002) Involvement of protein kinase C in fluoride induce apoptosis in different types of lung cells. *Ann NY Acad Sci* 973:218–220
59. Xu H, Jin XQ, Jing L, Li GS (2006) Effect of sodium fluoride on the expression of b-2 family and osteopontin in rat renal tubular cells. *Biol Trace Elem Res* 109:55–60
60. Zhan XA, Wang M, Xu ZR, Li WF, Li JX (2006) Evaluation of caspase dependent apoptosis during fluoride induced liver lesion in pigs. *Arch Toxicol* 80:74–80
61. Machalinska A (2001) NaF induced apoptosis in human bone marrow and cord blood CD34 positive cells. *Fluoride* 34:258–263
62. Chen J, Chen X, Yang K, Xia T, Xia H (2002) Studies on DNA damage and apoptosis in rat brain induced by fluoride. *Chin J Prevent Med* 36:222–224
63. Lu XH, Li GS, Sun B (2000) Study of the mechanism of neuron apoptosis in rats from the chronic fluorosis. *Chin J End* 19:96–98
64. Ge Y, Ning H, Feng C, Wang H, Yan X, Wang S, Wang J (2006) Apoptosis in brain cells of offspring rats exposed to high fluoride and low iodine. *Fluoride* 39:173–178
65. Shah SD, Chinoy NJ (2004) Adverse effects of fluoride and/or arsenic on the cerebral hemisphere of mice and recovery by some antidotes. *Fluoride* 37:162–171
66. Blaylock RL (2007) Fluoride neurotoxicity and excitotoxicity/microglial activation: critical need for more research. *Fluoride* 40:89–92
67. Blaylock RL (2004) Excitotoxicity: a possible central mechanism in fluoride neurotoxicity. *Fluoride* 37:301–314
68. Kumar A, Susheela AK (1994) Ultrastructural studies of spermiogenesis in rabbit exposed to chronic fluoride toxicity. *Int J Fert Menop Stud* 39:164–171
69. Pushpalatha T, Srinivas M, Sreenivasula Reddy P (2005) Exposure to high fluoride concentration in drinking water will affect spermatogenesis and steroidogenesis in male albino rats. *Bio-metals* 18:207–212
70. Chinoy NJ, Sequeira E (1989) Effects of fluoride on the histo-architecture of reproductive organs of the male mouse. *Reprod Toxicol* 3(4):261–267
71. Elbetieha A, Darmani H, Al-hiyasat AS (2000) Fertility effects of sodium fluoride in male mice. *Fluoride* 33:128–134
72. Huang C, Nui RY, Wang JD (2007) Toxic effect of sodium fluoride on reproductive function in male mice. *Fluoride* 40:162–168
73. Kumar N, Sood S, Arora B, Singh M, Beena (2010) Effect of duration of fluoride exposure on the reproductive system in male rabbits. *J Hum Rep Sci* 3:148–152
74. Long H, Jin Y, Lin M, Sun Y, Zhang L, Clinch C (2009) Fluoride toxicity in the male reproductive system. *Fluoride* 42(4):260–276
75. Tao S, Suttie JW (1976) Evidence for lack of an effect of dietary fluoride level on reproduction of mice. *J Nutr* 106:1115–1122
76. Collins TFX, Sprando RL, Black TN et al (2001) Multigenerational evaluation of sodium fluoride in rats. *Food Chem Toxicol* 39:601–612
77. Cui LX, Jiang CX, Wang XL (2003) Experimental study on effect of fluoride on reproductive system of male rats. *Chin J Endemiol* 22:195–197
78. Helal M, Dakdoky ME (2006) Fetotoxicity of fluoride in rats and the protective action of some antioxidants. *Fluoride* 39:202–210
79. Williams C, Berberof-Van Sane J, Dumont JE (1972) Inhibition of thyroid secretion by sodium fluoride in vitro. *Biophys Acta* 264:197–204
80. Hillman D, Bolkenbaugh DL, Convey EM (1979) Hypothyroidism and anaemia related to fluoride in dairy cattle. *J Dairy Sci* 62:265–277
81. Marasli N (1992) Studies in serum tri-iodothyronin (T_3) and thyroxine (T_4) levels in normal sheep and sheep with fluorosis. *Vet Facult Dergi Univ Ankara* 39:207–214
82. Bouaziz H, Soussia L, Guermazi F, Zeghal N (2005) Fluoride induced thyroid proliferative changes and their reversal in female mice and their pups. *Fluoride* 38:185–192
83. Kapoor V, Prasad T, Paliwal VK (2001) Blood biochemical constituents in calves following subclinical levels of fluoride toxicosis. *Fluoride* 34:126–131
84. Guan ZZ, Zhuang ZJ, Yang PS, Pan S (1988) Synergistic action of iodine deficiency and fluoride intoxication on rat thyroid. *Chin Med J* 101:679–684
85. Zhao W, Zhu H, Yu Z, Aoki K, Misumi J, Zhang X (1998) Long-term effects of various iodine and fluorine doses on the thyroid and fluorosis in mice. *Endocr Regul* 32:63–70
86. Bouaziz H, Ammar E, Ghorbel H et al (2004) Effect of fluoride ingested by lactating mice on the thyroid functions and bone maturation of their suckling pups. *Fluoride* 37(2):133–142
87. Trabelsi M, Guermazi F, Zeghal N (2001) Effect of fluoride on thyroid function and cerebellar development in mice. *Fluoride* 34:165–173
88. Rao K, Susheela AK (1979) Effect of sodium fluoride on adrenal gland of rabbit I. Studies on ascorbic acid and delta-5-3-beta hydroxyl steroid dehydrogenase activity. *Fluoride* 12:65–71
89. Zhang Y, Sun X, Sun G, Liu S, Wang L (2006) DNA damage induced by fluoride in rat osteoblasts. *Fluoride* 39:191–194
90. Podder S, Chattopadhyay A, Bhattacharya S, Ray MR (2008) Differential in vivo genotoxic effects of lower and higher concentrations of fluoride in mouse bone marrow cells. *Fluoride* 41:301–307

91. Khalil AM (1995) Chromosome aberrations in cultured rat bone marrow cells treated with inorganic fluorides. *Mut Res* 343:67–74
92. He LF, Chen JG (2006) DNA damage, apoptosis and cell cycle changes induced by fluoride in rat oral mucosal cells and hepatocytes. *World J Gastroenterol* 12:1144–1148
93. Shashi A (1993) Nucleic acid levels in thyroid gland in acute and chronic fluoride intoxication. *Fluoride* 29:191–196
94. Tripathi N, Bajpai S, Tripathi M (2009) Genotoxic alterations induced by fluoride in Asian catfish, *Clarias batrachus* (Linn.). *Fluoride* 42:292–296
95. Chinoy NJ, Shah SD (2004) Beneficial effects of some antidotes in fluoride and arsenic induced toxicity in kidney of mice. *Fluoride* 37:151–161
96. Khandare AL, Kumar PV, Lakshmaiah N (2000) Beneficial effect of tamarind ingestion on fluoride toxicity in dog. *Fluoride* 33:33–38
97. Ekambaram P, Namitha T, Bhuvaneswari S (2010) Therapeutic efficacy of *Tamarindus indica* (L) to protect against fluoride-induced oxidative stress in the liver of female rats. *Fluoride* 43:134–140
98. Trivedi MH, Verma RJ, Chinoy NJ (2009) Mitigation of sodium fluoride induced toxicity in mice brain by black tea infusion. *Fluoride* 42:29–33
99. Yen X, Chen L, Sun L et al (2010) Why do penguin not develop skeletal fluorosis? *Fluoride* 43:108–118